



12085CH03

Unit

2

# Electrochemistry

## Objectives

After studying this Unit, you will be able to

- describe an electrochemical cell and differentiate between galvanic and electrolytic cells;
- apply Nernst equation for calculating the emf of galvanic cell and define standard potential of the cell;
- derive relation between standard potential of the cell, Gibbs energy of cell reaction and its equilibrium constant;
- define resistivity ( $\rho$ ), conductivity ( $\kappa$ ) and molar conductivity ( $\Lambda_m$ ) of ionic solutions;
- differentiate between ionic (electrolytic) and electronic conductivity;
- describe the method for measurement of conductivity of electrolytic solutions and calculation of their molar conductivity;
- justify the variation of conductivity and molar conductivity of solutions with change in their concentration and define  $\Lambda_m^\circ$  (molar conductivity at zero concentration or infinite dilution);
- enunciate Kohlrausch law and learn its applications;
- understand quantitative aspects of electrolysis;
- describe the construction of some primary and secondary batteries and fuel cells;
- explain corrosion as an electrochemical process.

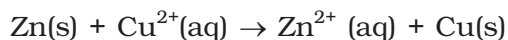
*Chemical reactions can be used to produce electrical energy, conversely, electrical energy can be used to carry out chemical reactions that do not proceed spontaneously.*

Electrochemistry is the study of production of electricity from energy released during spontaneous chemical reactions and the use of electrical energy to bring about non-spontaneous chemical transformations. The subject is of importance both for theoretical and practical considerations. A large number of metals, sodium hydroxide, chlorine, fluorine and many other chemicals are produced by electrochemical methods. Batteries and fuel cells convert chemical energy into electrical energy and are used on a large scale in various instruments and devices. The reactions carried out electrochemically can be energy efficient and less polluting. Therefore, study of electrochemistry is important for creating new technologies that are ecofriendly. The transmission of sensory signals through cells to brain and vice versa and communication between the cells are known to have electrochemical origin. Electrochemistry, is therefore, a very vast and interdisciplinary subject. In this Unit, we will cover only some of its important elementary aspects.

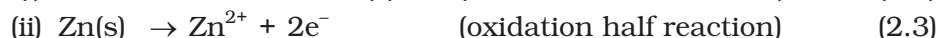
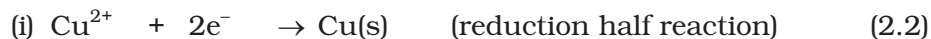
## 2.2 Galvanic Cells

As mentioned earlier a galvanic cell is an electrochemical cell that converts the chemical energy of a spontaneous redox reaction into electrical energy. In this device the **Gibbs energy** of the spontaneous redox reaction is converted into electrical work which may be used for running a motor or other electrical gadgets like heater, fan, geyser, etc.

Daniell cell discussed earlier is one such cell in which the following redox reaction occurs.



This reaction is a combination of two half reactions whose addition gives the overall cell reaction:



These reactions occur in two different portions of the Daniell cell. The reduction half reaction occurs on the copper electrode while the oxidation half reaction occurs on the zinc electrode. These two portions of the cell are also called **half-cells** or **redox couples**. The copper electrode may be called the reduction half cell and the zinc electrode, the oxidation half-cell.

We can construct innumerable number of galvanic cells on the pattern of Daniell cell by taking combinations of different half-cells. Each half-cell consists of a metallic electrode dipped into an electrolyte. The two half-cells are connected by a metallic wire through a voltmeter and a switch externally. The electrolytes of the two half-cells are connected internally through a salt bridge as shown in Fig. 2.1. Sometimes, both the electrodes dip in the same electrolyte solution and in such cases we do not require a salt bridge.

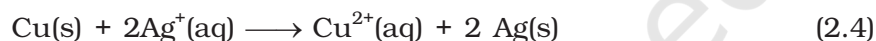
At each electrode-electrolyte interface there is a tendency of metal ions from the solution to deposit on the metal electrode trying to make it positively charged. At the same time, metal atoms of the electrode have a tendency to go into the solution as ions and leave behind the electrons at the electrode trying to make it negatively charged. At equilibrium, there is a separation of charges and depending on the tendencies of the two opposing reactions, the electrode may be positively or negatively charged with respect to the solution. A potential difference develops between the electrode and the electrolyte which is called **electrode potential**. When the concentrations of all the species involved in a half-cell is unity then the electrode potential is known as **standard electrode potential**. According to IUPAC convention, standard reduction potentials are now called standard electrode potentials. In a galvanic cell, the half-cell in which oxidation takes place is called **anode** and it has a negative potential with respect to the solution. The other half-cell in which reduction takes place is called **cathode** and it has a positive potential with respect to the solution. Thus, there exists a potential difference between the two electrodes and as soon as the switch is in the *on* position the electrons flow from negative electrode to positive electrode. The direction of current flow is opposite to that of electron flow.

The potential difference between the two electrodes of a galvanic cell is called the *cell potential* and is measured in volts. The **cell potential** is the difference between the electrode potentials (reduction potentials) of the cathode and anode. It is called the **cell electromotive force (emf)** of the cell when no current is drawn through the cell. It is now an accepted convention that we keep the anode on the left and the cathode on the right while representing the galvanic cell. A galvanic cell is generally represented by putting a vertical line between metal and electrolyte solution and putting a double vertical line between the two electrolytes connected by a salt bridge. Under this convention the emf of the cell is positive and is given by the potential of the half-cell on the right hand side minus the potential of the half-cell on the left hand side i.e.,

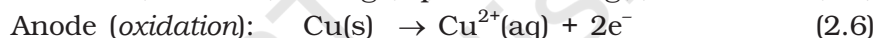
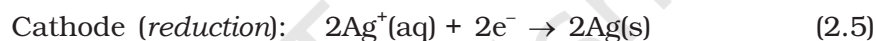
$$E_{\text{cell}} = E_{\text{right}} - E_{\text{left}}$$

This is illustrated by the following example:

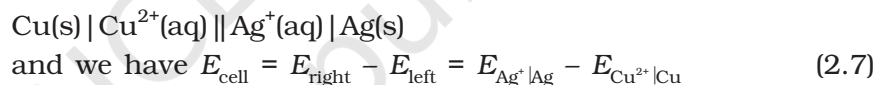
Cell reaction:



Half-cell reactions:



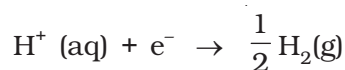
It can be seen that the sum of (3.5) and (3.6) leads to overall reaction (2.4) in the cell and that silver electrode acts as a cathode and copper electrode acts as an anode. The cell can be represented as:



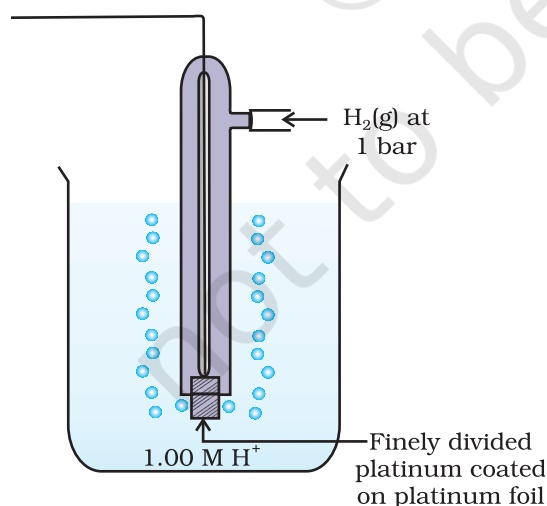
### 2.2.1 Measurement of Electrode Potential

The potential of individual half-cell cannot be measured. We can measure only the difference between the two half-cell potentials that gives the emf of the cell. If we arbitrarily choose the potential of one electrode (half-cell) then that of the other can be determined with respect

to this. According to convention, a half-cell called standard hydrogen electrode (Fig.3.3) represented by  $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq})$ , is assigned a zero potential at all temperatures corresponding to the reaction



The standard hydrogen electrode consists of a platinum electrode coated with platinum black. The electrode is dipped in an acidic solution and pure hydrogen gas is bubbled through it. The concentration of both the reduced and oxidised forms of hydrogen is maintained at unity (Fig. 2.3). This implies that the pressure of hydrogen gas is one bar and the concentration of hydrogen ion in the solution is one molar.



**Fig. 2.3:** Standard Hydrogen Electrode (SHE).

At 298 K the emf of the cell, standard hydrogen electrode || second half-cell constructed by taking standard hydrogen electrode as anode (reference half-cell) and the other half-cell as cathode, gives the reduction potential of the other half-cell. If the concentrations of the oxidised and the reduced forms of the species in the right hand half-cell are unity, then the cell potential is equal to standard electrode potential,  $E^\circ_R$  of the given half-cell.

$$E^\circ = E^\circ_R - E^\circ_L$$

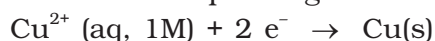
As  $E^\circ_L$  for standard hydrogen electrode is zero.

$$E^\circ = E^\circ_R - 0 = E^\circ_R$$

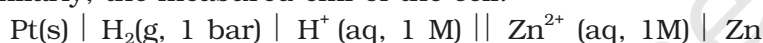
The measured emf of the cell:



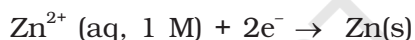
is 0.34 V and it is also the value for the standard electrode potential of the half-cell corresponding to the reaction:



Similarly, the measured emf of the cell:

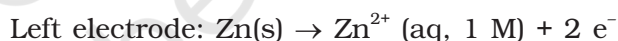


is -0.76 V corresponding to the standard electrode potential of the half-cell reaction:

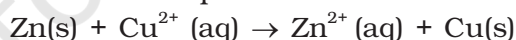


The positive value of the standard electrode potential in the first case indicates that  $\text{Cu}^{2+}$  ions get reduced more easily than  $\text{H}^+$  ions. The reverse process cannot occur, that is, hydrogen ions cannot oxidise Cu (or alternatively we can say that hydrogen gas can reduce copper ion) under the standard conditions described above. Thus, Cu does not dissolve in HCl. In nitric acid it is oxidised by nitrate ion and not by hydrogen ion. The negative value of the standard electrode potential in the second case indicates that hydrogen ions can oxidise zinc (or zinc can reduce hydrogen ions).

In view of this convention, the half reaction for the Daniell cell in Fig. 2.1 can be written as:

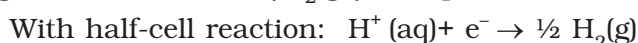


The overall reaction of the cell is the sum of above two reactions and we obtain the equation:



$$\begin{aligned} \text{emf of the cell} &= E^\circ_{\text{cell}} = E^\circ_R - E^\circ_L \\ &= 0.34 \text{ V} - (-0.76) \text{ V} = 1.10 \text{ V} \end{aligned}$$

Sometimes metals like platinum or gold are used as inert electrodes. They do not participate in the reaction but provide their surface for oxidation or reduction reactions and for the conduction of electrons. For example, Pt is used in the following half-cells:



**Example 2.2** Calculate the equilibrium constant of the reaction:  
 $\text{Cu(s)} + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag(s)}$

$$E_{(\text{cell})}^{\circ} = 0.46 \text{ V}$$

**Solution**  $E_{(\text{cell})}^{\circ} = \frac{0.059 \text{ V}}{2} \log K_C = 0.46 \text{ V}$  or

$$\log K_C = \frac{0.46 \text{ V} \times 2}{0.059 \text{ V}} = 15.6$$

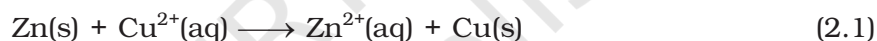
$$K_C = 3.92 \times 10^{15}$$

### 2.3.2 Electro-chemical Cell and Gibbs Energy of the Reaction

Electrical work done in one second is equal to electrical potential multiplied by total charge passed. If we want to obtain maximum work from a galvanic cell then charge has to be passed reversibly. The reversible work done by a galvanic cell is equal to decrease in its Gibbs energy and therefore, if the emf of the cell is  $E$  and  $nF$  is the amount of charge passed and  $\Delta_r G$  is the Gibbs energy of the reaction, then

$$\Delta_r G = - nFE_{(\text{cell})} \quad (2.15)$$

It may be remembered that  $E_{(\text{cell})}$  is an intensive parameter but  $\Delta_r G$  is an extensive thermodynamic property and the value depends on  $n$ . Thus, if we write the reaction



$$\Delta_r G = - 2FE_{(\text{cell})}$$

but when we write the reaction



$$\Delta_r G = - 4FE_{(\text{cell})}$$

If the concentration of all the reacting species is unity, then

$$E_{(\text{cell})} = E_{(\text{cell})}^{\circ} \text{ and we have}$$

$$\Delta_r G^{\circ} = - nFE_{(\text{cell})}^{\circ} \quad (2.16)$$

Thus, from the measurement of  $E_{(\text{cell})}^{\circ}$  we can obtain an important thermodynamic quantity,  $\Delta_r G^{\circ}$ , standard Gibbs energy of the reaction. From the latter we can calculate equilibrium constant by the equation:

$$\Delta_r G^{\circ} = -RT \ln K.$$

**Example 2.3** The standard electrode potential for Daniell cell is 1.1V. Calculate the standard Gibbs energy for the reaction:



**Solution**  $\Delta_r G^{\circ} = - nFE_{(\text{cell})}^{\circ}$

$n$  in the above equation is 2,  $F = 96487 \text{ C mol}^{-1}$  and  $E_{(\text{cell})}^{\circ} = 1.1 \text{ V}$

$$\begin{aligned} \text{Therefore, } \Delta_r G^{\circ} &= - 2 \times 1.1\text{V} \times 96487 \text{ C mol}^{-1} \\ &= - 21227 \text{ J mol}^{-1} \\ &= - 212.27 \text{ kJ mol}^{-1} \end{aligned}$$

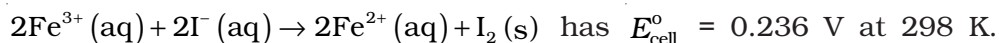
**2.4** Calculate the potential of hydrogen electrode in contact with a solution whose pH is 10.

**2.5** Calculate the emf of the cell in which the following reaction takes place:



Given that  $E_{\text{cell}}^{\circ} = 1.05 \text{ V}$

**2.6** The cell in which the following reaction occurs:



Calculate the standard Gibbs energy and the equilibrium constant of the cell reaction.

## 2.4 Conductance of Electrolytic Solutions

It is necessary to define a few terms before we consider the subject of conductance of electricity through electrolytic solutions. The electrical resistance is represented by the symbol 'R' and it is measured in ohm ( $\Omega$ ) which in terms of SI base units is equal to  $(\text{kg m}^2)/(\text{S}^3 \text{ A}^2)$ . It can be measured with the help of a **Wheatstone bridge** with which you are familiar from your study of physics. The electrical resistance of any object is directly proportional to its length,  $l$ , and inversely proportional to its area of cross section,  $A$ . That is,

$$R \propto \frac{l}{A} \text{ or } R = \rho \frac{l}{A} \quad (2.17)$$

The constant of proportionality,  $\rho$  (Greek, rho), is called **resistivity** (specific resistance). Its SI units are ohm metre ( $\Omega \text{ m}$ ) and quite often its submultiple, ohm centimetre ( $\Omega \text{ cm}$ ) is also used. IUPAC recommends the use of the term resistivity over specific resistance and hence in the rest of the book we shall use the term resistivity. Physically, the resistivity for a substance is its resistance when it is one metre long and its area of cross section is one  $\text{m}^2$ . It can be seen that:

$$1 \Omega \text{ m} = 100 \Omega \text{ cm} \text{ or } 1 \Omega \text{ cm} = 0.01 \Omega \text{ m}$$

The inverse of resistance,  $R$ , is called **conductance**,  $G$ , and we have the relation:

$$G = \frac{1}{R} = \frac{A}{\rho l} = \kappa \frac{A}{l} \quad (2.18)$$

The SI unit of conductance is siemens, represented by the symbol 'S' and is equal to  $\text{ohm}^{-1}$  (also known as mho) or  $\Omega^{-1}$ . The inverse of resistivity, called **conductivity** (specific conductance) is represented by the symbol,  $\kappa$  (Greek, kappa). IUPAC has recommended the use of term conductivity over specific conductance and hence we shall use the term conductivity in the rest of the book. The SI units of conductivity are  $\text{S m}^{-1}$  but quite often,  $\kappa$  is expressed in  $\text{S cm}^{-1}$ . Conductivity of a material in  $\text{S m}^{-1}$  is its conductance when it is 1 m long and its area of cross section is  $1 \text{ m}^2$ . It may be noted that  $1 \text{ S cm}^{-1} = 100 \text{ S m}^{-1}$ .



As the electrons enter at one end and go out through the other end, the composition of the metallic conductor remains unchanged. The mechanism of conductance through semiconductors is more complex.

We already know that even very pure water has small amounts of hydrogen and hydroxyl ions ( $\sim 10^{-7}\text{M}$ ) which lend it very low conductivity ( $3.5 \times 10^{-5} \text{ S m}^{-1}$ ). When electrolytes are dissolved in water, they furnish their own ions in the solution hence its conductivity also increases. The conductance of electricity by ions present in the solutions is called electrolytic or **ionic conductance**. The conductivity of electrolytic (ionic) solutions depends on:

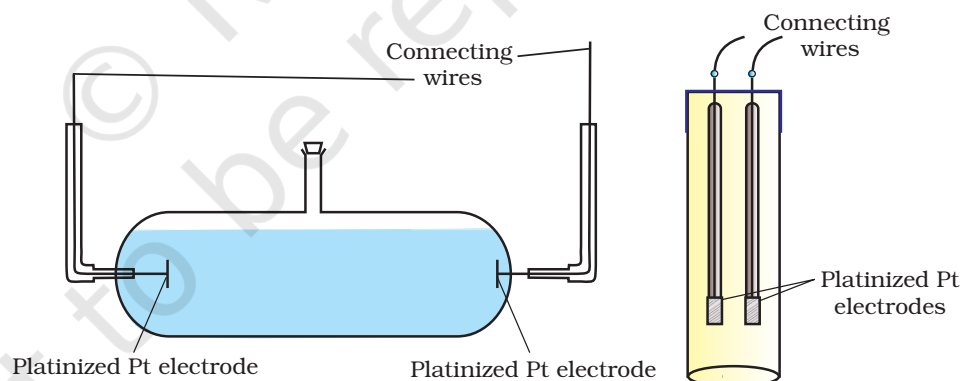
- (i) the nature of the electrolyte added
- (ii) size of the ions produced and their solvation
- (iii) the nature of the solvent and its viscosity
- (iv) concentration of the electrolyte
- (v) temperature (it increases with the increase of temperature).

Passage of direct current through ionic solution over a prolonged period can lead to change in its composition due to electrochemical reactions (Section 2.4.1).

#### 2.4.1 Measurement of the Conductivity of Ionic Solutions

We know that accurate measurement of an unknown resistance can be performed on a Wheatstone bridge. However, for measuring the resistance of an ionic solution we face two problems. Firstly, passing direct current (DC) changes the composition of the solution. Secondly, a solution cannot be connected to the bridge like a metallic wire or other solid conductor. The first difficulty is resolved by using an alternating current (AC) source of power. The second problem is solved by using a specially designed vessel called **conductivity cell**. It is available in several designs and two simple ones are shown in Fig. 2.4.

**Fig. 2.4**  
Two different types of conductivity cells.



Basically it consists of two platinum electrodes coated with platinum black (finely divided metallic Pt is deposited on the electrodes electrochemically). These have area of cross section equal to 'A' and are separated by distance 'l'. Therefore, solution confined between these electrodes is a column of length l and area of cross section A. The resistance of such a column of solution is then given by the equation:

$$R = \rho \frac{l}{A} = \frac{l}{\kappa A} \quad (2.17)$$